CHEMISTRY III

Chemical properties of alcohols, and identification of aldehydes and ketones

Professor: _____ Date: _____ Group: _____

Student's name	Registration
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Pre-laboratory activity

1. Take the Canvas quiz.

Objectives

- Identify three different types of alcohols and their classification using various techniques.
- Identify the carbonyl group in compounds such as aldehydes and ketones by the DNPH (2,4-dinitrophenylhydrazine) test.
- Distinguish between aldehydes and ketones by the Tollens reaction.

Materials and reagents

Materials

- 12 test tubes (20 x 150)
- Test tube clamps
- Test tube forceps
- Pasteur pipettes

Reagents

- Ethanol or methanol
- Isopropanol (2-propanol)
- Tert-butanol
- Acetone
- Formaldehyde
- Benzaldehyde
- Potassium permanganate solution
- Tollens reagent
- Chromic acid

Precautions and emergency measures

Safety precautions for all reagents: wear a white cotton lab coat with long sleeves, protective gloves and safety glasses; use a ventilator and/or gas extraction system or airway protection.

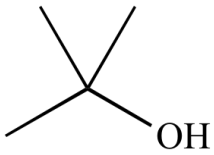
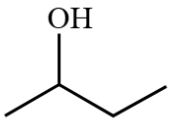
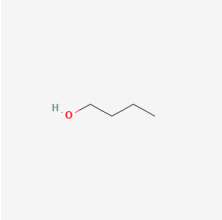
Reagent	Actions in case of emergency	Notes
Ethanol / isopropanol / tert-butanol	Inhalation: move to fresh air; give artificial respiration and oxygen if breathing is difficult or absent. Skin: remove contaminated clothing, wash with soap and water. Eyes: flush with plenty of water, remove contact lenses, call an ophthalmologist. Ingestion: do not induce vomiting.	Flammable
Acetone	Inhalation: move to fresh air; artificial respiration and oxygen if needed. Skin: remove contaminated clothing, wash with soap and water. Eyes: remove contact lenses, rinse with water for at least 10 minutes, call a physician. Ingestion: rinse mouth, do not induce vomiting.	Flammable; irritant
Formaldehyde	Inhalation: move to a ventilated area; artificial respiration and oxygen if needed. Skin: remove contaminated clothing, wash with soap and water for at least 20 minutes. Eyes: remove contact lenses, wash with water for at least 10 minutes lifting eyelids. Ingestion: rinse mouth, do not induce vomiting; drink a glass of water if the patient allows it.	Toxic; environmental hazard; flammable
Chromic acid	Inhalation: move to a ventilated area; artificial respiration and oxygen if needed. Skin: remove contaminated clothing, wash with soap and water for at least 20 minutes. Eyes: remove contact lenses, wash with water for at least 10 minutes lifting eyelids. Ingestion: rinse mouth, do not induce vomiting; drink a glass of water if the patient allows it.	Toxic; environmental hazard; health hazard; irritant

Part A: Identification of primary, secondary and tertiary alcohols

1. You will receive samples of 3 alcohols (ethanol, 2-propanol and tert-butanol, not necessarily in that order), labeled 1, 2 and 3, not necessarily in the order mentioned.
2. To each tube of the series labeled 1, 2, 3, add approximately 4 drops of potassium permanganate solution and let the sample react, if necessary. Observe whether some samples react faster than others.
3. To another series of labeled tubes, add approximately 4 drops of chromic acid and wait about 3 minutes for the reactions to complete. As before, note any difference in the speed of reaction.

4. Once the samples have finished reacting, write your observations in the table below. According to the order of reaction of the alcohols, identify the type of alcohol (primary, secondary or tertiary) and write its formula.

Order of reactivity of alcohols: Primary > Secondary > Tertiary

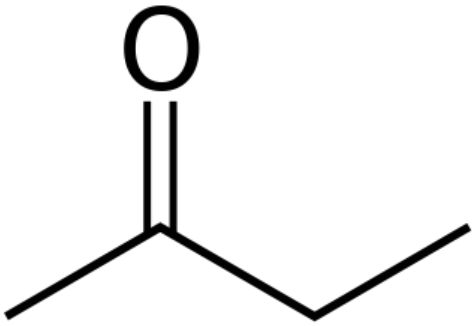
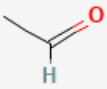
Substance	Reacted with potassium permanganate? YES/NO. Assign 1 to the fastest and 3 to the slowest.	Reacted with chromic acid? YES/NO. Assign 1 to the fastest and 3 to the slowest.	Name and type of alcohol (primary, secondary, tertiary)	Zigzag formula of the molecule
1 -- 1 ml	NO	YES (reacted extremely fast with a greenish precipitate	3rd	
2 – 1 ml	YES	NO (dark green mixture)	2nd	
3 – 1 ml	YES	YES (extremely fast with a greenish precipitate)	1st	

Part C: Tollens reagent test to identify aldehydes and ketones

1. Take the test tubes labeled 4 and 5, add 2 mL of Tollens reagent to each, and observe what happens.
2. If a silver mirror appears, the compound is an aldehyde; otherwise, it is a ketone.

Using the results of Parts B and C, identify substances 4 and 5 by assigning the name and writing the formula. Remember that one is an aldehyde and the other is acetone.

Results – Table 1

Substance	Name	Zigzag or semi-condensed formula
4 – 1 ml	Ketone	
5 – 1 ml	Aldehyde	

Report

1. Write at least one comprehensible conclusion about your work in the lab (minimum 120 words). Clearly state the purpose of the session and whether it was accomplished.
2. The report must include a cover page with the names and registration numbers of the members. The body must contain a brief introduction, objectives, answers to the research prior to the experimentation, procedures, results (tables), conclusion, and references in APA format, taking care of spelling and presentation. Do not include the instructions of this practice.
3. Hand in your report to the professor in the time and form indicated. As of the second midterm, late practices and practices without the requested format will no longer be accepted.

Based on the recorded results, the alcohols showed different reactions when exposed to potassium permanganate and chromic acid. These differences helped classify the substances according to their reactivity. The results generally supported the expected order of reactivity: primary alcohols react more easily than secondary ones, while tertiaries are more resistant to oxidation.

Additionally, the identification of aldehydes and ketones was based on their different chemical propperties, particular